Progressive Education Society'sModern College of Engineering, Pune-05.(An Autonomous Institute Affiliated to Savitribai Phule Pune University) Department of S.Y. B. Tech [AIDS]

Assignment No: 09

SUBJECT: Java Programming

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Class: SY BTech (AIDS)

Section: A

Roll Number: 26040

Title / Problem Statement :

Develop an application to manage student information using a text file. The application should create a new file if it doesn't already exist, allow users to enter student details (such as name, id, and grade) with options to overwrite or append to the existing records, and ensure proper file handling by closing and reopening the file when needed. The application should also be able to read and display

Input :

```java
import java.io.*;
import java.util.*;

class Student {
    String name;
    int id;
    double grade;

    Student(String name, int id, double grade) {
        this.name = name;
        this.id = id;
        this.grade = grade;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        File file = new File("students.txt");

        try {
            // Create file if not exists
            if (!file.exists()) {
                file.createNewFile();
                System.out.println("File created.");
            }
```

```java
// Choose mode
System.out.print("Enter mode (append/overwrite): ");
String mode = sc.nextLine();

FileWriter fw;
if (mode.equalsIgnoreCase("append")) {
    fw = new FileWriter(file, true);
} else {
    fw = new FileWriter(file, false);
}

BufferedWriter bw = new BufferedWriter(fw);

// Input student details
System.out.print("Enter number of students: ");
int n = sc.nextInt();
sc.nextLine();

for (int i = 0; i < n; i++) {
    System.out.print("Enter name: ");
    String name = sc.nextLine();

    System.out.print("Enter ID: ");
    int id = sc.nextInt();

    System.out.print("Enter Grade: ");
    double grade = sc.nextDouble();
    sc.nextLine();
```

```
            bw.write(name + "," + id + "," + grade);
            bw.newLine();
        }

        bw.close();

        // Reading file
        BufferedReader br = new BufferedReader(new FileReader(file));
        List<Student> list = new ArrayList<>();

        String line;
        while ((line = br.readLine()) != null) {
            String[] data = line.split(",");
            list.add(new Student(data[0], Integer.parseInt(data[1]),
Double.parseDouble(data[2])));
        }

        br.close();

        // Sorting by name
        Collections.sort(list, (a, b) -> a.name.compareToIgnoreCase(b.name));

        // Display
        System.out.println("\n--- Sorted Student Records ---");
        for (Student s : list) {
            System.out.println("Name: " + s.name + ", ID: " + s.id + ", Grade: " +
s.grade);
        }
```

```
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output :

```
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_09$ javac Main.java
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_09$ java Main
File created.
Enter mode (append/overwrite): append
Enter number of students: 5
Enter name: Prasad
Enter ID: 7
Enter Grade: 10
Enter name: Saee
Enter ID: 8
Enter Grade: 10
Enter name: Ayush
Enter ID: 9
Enter Grade: 10
Enter name: Abhimanyu
Enter ID: 10
Enter Grade: 9
Enter name: Atharva
Enter ID: 11
Enter Grade: 8

--- Sorted Student Records ---
Name: Abhimanyu, ID: 10, Grade: 9.0
Name: Atharva, ID: 11, Grade: 8.0
Name: Ayush, ID: 9, Grade: 10.0
Name: Prasad, ID: 7, Grade: 10.0
Name: Saee, ID: 8, Grade: 10.0
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_09$
```